

ME404: Dynamics and Control of Mechanical Systems, Fall 2026

Problem Set #1: Modeling and Dynamics — Answer Key

Angles are in radians. Transfer functions use zero initial conditions unless stated otherwise. Each answer uses the coordinate and sign conventions in the problem statement.

Problem #1

(Part A) Dubin's Kinematics

The unit vector along the vehicle's heading is

$$e_{x_b} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}.$$

Because the cart has no sideways velocity, its inertial velocity is $v e_{x_b}$. Thus

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega.$$

Equivalently,

$$\dot{q} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}.$$

The no-slip constraint is $-\dot{x} \sin \theta + \dot{y} \cos \theta = 0$.

(Part B) Dubin's Dynamics

(a) Newton's and Euler's laws in the rotating body frame.

Write the velocity as $\mathbf{v} = v_x e_{x_b} + v_y e_{y_b}$. Since

$$\dot{e}_{x_b} = \omega e_{y_b}, \quad \dot{e}_{y_b} = -\omega e_{x_b},$$

the inertial acceleration is

$$\mathbf{a} = (\dot{v}_x - \omega v_y) e_{x_b} + (\dot{v}_y + \omega v_x) e_{y_b}.$$

Applying Newton's law along the body axes and Euler's law about the center of mass gives

$$m(\dot{v}_x - \omega v_y) = F, \quad m(\dot{v}_y + \omega v_x) = 0, \quad J\dot{\omega} = M.$$

Therefore,

$$\dot{v}_x = \frac{F}{m} + \omega v_y, \quad \dot{v}_y = -\omega v_x, \quad \dot{\omega} = \frac{M}{J}.$$

The velocity-coupling terms arise from the rotating coordinates, even though there is no applied lateral force.

(b) Full state model.

Rotate both body-frame velocity components into the inertial frame:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix}.$$

For $q = (x, y, \theta, v_x, v_y, \omega)^T$ and $u = (F, M)^T$,

$$\dot{q} = \begin{bmatrix} v_x \cos \theta - v_y \sin \theta \\ v_x \sin \theta + v_y \cos \theta \\ \omega \\ F/m + \omega v_y \\ -\omega v_x \\ M/J \end{bmatrix}.$$

Part A's position equations are recovered instantaneously when $v_y = 0$ and $v_x = v$. However, $v_y = 0$ generally does not remain true in this slipping model: if $\omega v_x \neq 0$, then $\dot{v}_y \neq 0$. Maintaining no slip during a turn would require a lateral constraint force.

Problem #2

For compactness, define the constant combinations

$$\begin{aligned} A &= J_1 + m_1 l_{c1}^2 + m_2 l_1^2, \\ D &= J_2 + m_2 l_{c2}^2, \\ B &= m_2 l_1 l_{c2}, \\ P &= (m_1 l_{c1} + m_2 l_1)g, \\ Q &= m_2 l_{c2}g. \end{aligned}$$

Here A, B, D have units of moment of inertia, and P, Q have units of torque. The symbol g alone denotes gravitational acceleration; $g(\theta)$ below denotes the gravity vector.

(a) Center-of-mass kinematics and energies

With O as the origin and the inertial y -axis upward,

$$r_{c1} = \begin{bmatrix} l_{c1} \cos \theta_1 \\ l_{c1} \sin \theta_1 \end{bmatrix}, \quad r_{c2} = \begin{bmatrix} l_1 \cos \theta_1 + l_{c2} \cos(\theta_1 + \theta_2) \\ l_1 \sin \theta_1 + l_{c2} \sin(\theta_1 + \theta_2) \end{bmatrix}.$$

Differentiation gives

$$\begin{aligned} \dot{r}_{c1} &= l_{c1} \dot{\theta}_1 \begin{bmatrix} -\sin \theta_1 \\ \cos \theta_1 \end{bmatrix}, \\ \dot{r}_{c2} &= l_1 \dot{\theta}_1 \begin{bmatrix} -\sin \theta_1 \\ \cos \theta_1 \end{bmatrix} + l_{c2} (\dot{\theta}_1 + \dot{\theta}_2) \begin{bmatrix} -\sin(\theta_1 + \theta_2) \\ \cos(\theta_1 + \theta_2) \end{bmatrix}. \end{aligned}$$

In particular,

$$\|\dot{r}_{c2}\|^2 = l_1^2 \dot{\theta}_1^2 + l_{c2}^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 + 2l_1 l_{c2} \cos \theta_2 \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2).$$

The absolute angular velocities of the links are $\dot{\theta}_1$ and $\dot{\theta}_1 + \dot{\theta}_2$. Consequently,

$$\begin{aligned} T &= \frac{1}{2} m_1 \|\dot{r}_{c1}\|^2 + \frac{1}{2} J_1 \dot{\theta}_1^2 + \frac{1}{2} m_2 \|\dot{r}_{c2}\|^2 + \frac{1}{2} J_2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \\ &= \boxed{\frac{1}{2} A \dot{\theta}_1^2 + \frac{1}{2} D (\dot{\theta}_1 + \dot{\theta}_2)^2 + B \cos \theta_2 \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2)}. \end{aligned}$$

Taking potential energy to be zero when the centers of mass are at $y = 0$,

$$V = P \sin \theta_1 + Q \sin(\theta_1 + \theta_2).$$

The parameter l_2 does not appear separately: its effect is already represented by l_{c2} and J_2 . No uniform-link assumption is needed.

(b) Nonlinear equations of motion

The virtual work of the motors is

$$\delta W = \tau_1 \delta \theta_1 + \tau_2 [\delta(\theta_1 + \theta_2) - \delta \theta_1] = \tau_1 \delta \theta_1 + \tau_2 \delta \theta_2.$$

Thus the generalized forces are $(\tau_1, \tau_2)^T$, including the elbow motor's reaction torque on link 1. Lagrange's equations are

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\theta}_i} - \frac{\partial T}{\partial \theta_i} + \frac{\partial V}{\partial \theta_i} = \tau_i.$$

Useful intermediate derivatives are

$$\begin{aligned} \frac{\partial T}{\partial \dot{\theta}_1} &= (A + D + 2B \cos \theta_2) \dot{\theta}_1 + (D + B \cos \theta_2) \dot{\theta}_2, \\ \frac{\partial T}{\partial \dot{\theta}_2} &= (D + B \cos \theta_2) \dot{\theta}_1 + D \dot{\theta}_2, \\ \frac{\partial T}{\partial \theta_1} &= 0, \quad \frac{\partial T}{\partial \theta_2} = -B \sin \theta_2 \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2). \end{aligned}$$

Substituting gives

$$\begin{aligned} (A + D + 2B \cos \theta_2) \ddot{\theta}_1 + (D + B \cos \theta_2) \ddot{\theta}_2 \\ - B \sin \theta_2 (2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) + P \cos \theta_1 + Q \cos(\theta_1 + \theta_2) &= \tau_1, \\ (D + B \cos \theta_2) \ddot{\theta}_1 + D \ddot{\theta}_2 + B \sin \theta_2 \dot{\theta}_1^2 + Q \cos(\theta_1 + \theta_2) &= \tau_2. \end{aligned}$$

One valid representation of $M\ddot{\theta} + C\dot{\theta} + g(\theta) = \tau$ is

$$M(\theta) = \begin{bmatrix} A + D + 2B \cos \theta_2 & D + B \cos \theta_2 \\ D + B \cos \theta_2 & D \end{bmatrix},$$

$$C(\theta, \dot{\theta}) = \begin{bmatrix} -B \sin \theta_2 \dot{\theta}_2 & -B \sin \theta_2 (\dot{\theta}_1 + \dot{\theta}_2) \\ B \sin \theta_2 \dot{\theta}_1 & 0 \end{bmatrix},$$

$$g(\theta) = \begin{bmatrix} P \cos \theta_1 + Q \cos(\theta_1 + \theta_2) \\ Q \cos(\theta_1 + \theta_2) \end{bmatrix}.$$

The matrix C is not unique; the velocity-force vector $C\dot{\theta}$ is the quantity fixed by the equations. The displayed choice also makes $\dot{M} - 2C$ skew-symmetric, as expected from the kinetic-energy balance.

(c) Nonlinearity and inertia coupling

The nonlinear terms are the configuration-dependent factors $\cos \theta_2$ multiplying accelerations, the quadratic velocity terms multiplied by $\sin \theta_2$, and the trigonometric gravity terms.

The off-diagonal entries represent inertial coupling: accelerating either joint generally requires torque at the other joint because both coordinates affect link 2's motion. In particular, changing θ_1 translates the elbow and rotates

link 2, while changing θ_2 changes link 2's motion relative to link 1.

Symmetry follows directly from

$$M_{12} = M_{21} = D + B \cos \theta_2.$$

It also follows because M is the Hessian of T with respect to the generalized velocities. As a useful check,

$$\det M = AD - B^2 \cos^2 \theta_2 \geq AD - B^2,$$

and expanding the lower bound gives

$$AD - B^2 = J_1 J_2 + J_1 m_2 l_{c2}^2 + J_2 m_2 l_1^2 + J_2 m_1 l_{c1}^2 + m_1 m_2 l_{c1}^2 l_{c2}^2 > 0.$$

The term $m_2^2 l_1^2 l_{c2}^2$ cancels exactly. For positive masses and link inertias $J_i > 0$, the remaining terms establish that $\det M > 0$ at every configuration. Together with $M_{22} = D > 0$, this proves that M is positive definite.

(d) Linearization about the upward equilibrium

At $\theta_1 = \pi/2$, $\theta_2 = 0$, and $\dot{\theta} = 0$, both gravity components vanish. With $\tau = 0$, the equations therefore permit $\ddot{\theta} = 0$: this is an equilibrium.

Let $\varphi_1 = \theta_1 - \pi/2$ and $\varphi_2 = \theta_2$. To first order,

$$\cos \theta_2 \approx 1, \quad \cos \theta_1 \approx -\varphi_1, \quad \cos(\theta_1 + \theta_2) \approx -(\varphi_1 + \varphi_2).$$

Hence

$$M_0 \ddot{\varphi} + K \varphi = \tau,$$

where

$$M_0 = \begin{bmatrix} A + D + 2B & D + B \\ D + B & D \end{bmatrix} = \begin{bmatrix} J_1 + m_1 l_{c1}^2 + J_2 + m_2 (l_1 + l_{c2})^2 & J_2 + m_2 l_{c2} (l_1 + l_{c2}) \\ J_2 + m_2 l_{c2} (l_1 + l_{c2}) & J_2 + m_2 l_{c2}^2 \end{bmatrix},$$

$$K = - \begin{bmatrix} P + Q & Q \\ Q & Q \end{bmatrix} = -g \begin{bmatrix} m_1 l_{c1} + m_2 l_1 + m_2 l_{c2} & m_2 l_{c2} \\ m_2 l_{c2} & m_2 l_{c2} \end{bmatrix}.$$

There is no first-order contribution from $C\dot{\theta}$: it is quadratic in velocities about a rest equilibrium. At this particular equilibrium, the additional factor $\sin \theta_2 \approx \varphi_2$ makes its leading contribution third order in small deviations. Variations of M also contribute no linear correction beyond $M_0 \ddot{\varphi}$.

The negative stiffness is physically appropriate: gravity drives a small displacement away from the upward configuration. For $P, Q > 0$, K is negative definite. Since M_0 is positive definite, $M_0^{-1}(-K)$ has positive eigenvalues μ_i , giving linearized poles $s = \pm \sqrt{\mu_i}$. The positive real poles establish instability.

(e) Four transfer functions and characteristic polynomial

The outputs are the angular deviations φ_1, φ_2 , and the inputs are τ_1, τ_2 . With zero initial deviations and velocities,

$$\begin{bmatrix} (A + D + 2B)s^2 - (P + Q) & (D + B)s^2 - Q \\ (D + B)s^2 - Q & Ds^2 - Q \end{bmatrix} \begin{bmatrix} \Phi_1 \\ \Phi_2 \end{bmatrix} = \begin{bmatrix} \mathcal{T}_1 \\ \mathcal{T}_2 \end{bmatrix}.$$

The common characteristic polynomial is

$$\begin{aligned}
\chi(s) &= \det(M_0 s^2 + K) \\
&= [(A + D + 2B)s^2 - (P + Q)](Ds^2 - Q) - [(D + B)s^2 - Q]^2 \\
&= \boxed{(AD - B^2)s^4 - (DP + AQ)s^2 + PQ}.
\end{aligned}$$

The monic state-space characteristic polynomial is $\chi(s)/(AD - B^2)$. Inverting the 2×2 matrix gives all four transfer functions, with the other input set to zero in each ratio:

$$\begin{aligned}
\boxed{\frac{\Phi_1}{\mathcal{T}_1} = \frac{Ds^2 - Q}{\chi(s)}, \quad \frac{\Phi_1}{\mathcal{T}_2} = \frac{Q - (D + B)s^2}{\chi(s)},} \\
\boxed{\frac{\Phi_2}{\mathcal{T}_1} = \frac{Q - (D + B)s^2}{\chi(s)}, \quad \frac{\Phi_2}{\mathcal{T}_2} = \frac{(A + D + 2B)s^2 - (P + Q)}{\chi(s)}}.
\end{aligned}$$

The equality of the cross transfer functions follows from the symmetric mass and stiffness matrices. To locate the poles, treat $\chi(s)$ as a quadratic in s^2 . Its discriminant is

$$(DP + AQ)^2 - 4(AD - B^2)PQ = (DP - AQ)^2 + 4B^2PQ > 0$$

for $B, P, Q > 0$. The two roots in s^2 are therefore real and distinct. Their sum $(DP + AQ)/(AD - B^2)$ and product $PQ/(AD - B^2)$ are positive, so both roots are positive. Thus the poles form two distinct real pairs $\pm\lambda_1, \pm\lambda_2$, with $\lambda_i > 0$, consistent with the upward equilibrium's instability.

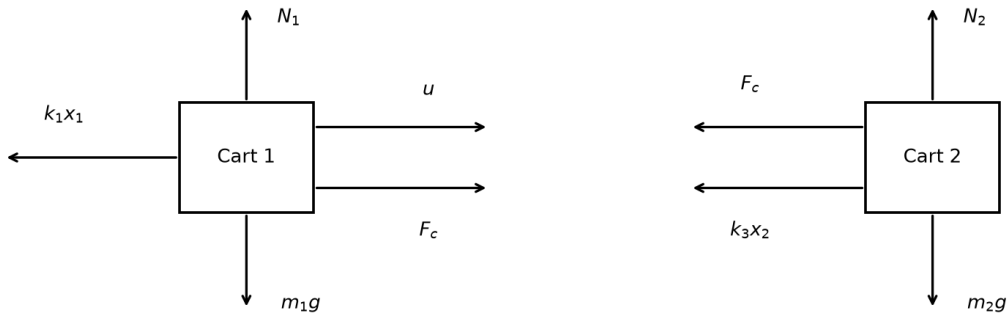
Problem #3

(a) Free-body diagrams and equations of motion

Define the signed coupling force on cart 1 by

$$F_c = k_2(x_2 - x_1) + b(\dot{x}_2 - \dot{x}_1).$$

Cart 2 experiences the opposite force, $-F_c$. The arrows below indicate positive directions for the labeled signed quantities; a negative value reverses the force. The vertical forces satisfy $N_i = m_i g$.



Horizontal force balance gives

$$\begin{aligned}
m_1 \ddot{x}_1 &= u - k_1 x_1 + k_2(x_2 - x_1) + b(\dot{x}_2 - \dot{x}_1), \\
m_2 \ddot{x}_2 &= -k_3 x_2 - k_2(x_2 - x_1) - b(\dot{x}_2 - \dot{x}_1).
\end{aligned}$$

Equivalently,

$$\begin{aligned} m_1 \ddot{x}_1 + b(\dot{x}_1 - \dot{x}_2) + (k_1 + k_2)x_1 - k_2 x_2 &= u, \\ m_2 \ddot{x}_2 + b(\dot{x}_2 - \dot{x}_1) + (k_2 + k_3)x_2 - k_2 x_1 &= 0. \end{aligned}$$

The spring k_2 and damper b connect the two carts, so their equal-and-opposite forces enter both equations. Their extension and extension rate depend only on relative displacement and relative velocity, respectively; translating both carts equally does not deform either coupling element.

(b) Transfer functions, characteristic polynomial, and zeros

Define

$$A_1(s) = m_1 s^2 + bs + k_1 + k_2, \quad A_2(s) = m_2 s^2 + bs + k_2 + k_3, \quad H(s) = bs + k_2.$$

With zero initial conditions, the transformed equations are

$$\begin{bmatrix} A_1 & -H \\ -H & A_2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} U \\ 0 \end{bmatrix}.$$

Thus

$$\frac{X_1(s)}{U(s)} = \frac{A_2(s)}{\Delta(s)}, \quad \frac{X_2(s)}{U(s)} = \frac{H(s)}{\Delta(s)},$$

where

$$\begin{aligned} \Delta(s) &= A_1(s)A_2(s) - H(s)^2 \\ &= [m_1 m_2 s^4 + b(m_1 + m_2)s^3 + [m_1(k_2 + k_3) + m_2(k_1 + k_2)]s^2 + b(k_1 + k_3)s + k_1 k_2 + k_1 k_3 + k_2 k_3]. \end{aligned}$$

For generic parameters without pole-zero cancellations, the two finite zeros of X_1/U are

$$z_{1,\pm} = \frac{-b \pm \sqrt{b^2 - 4m_2(k_2 + k_3)}}{2m_2}.$$

This formula also applies at $b = 0$, when $z_{1,\pm} = \pm i\sqrt{(k_2 + k_3)/m_2}$. For $b > 0$, the single finite zero of X_2/U is

$$z_2 = -\frac{k_2}{b}.$$

The zeros of X_1/U are the roots of $A_2(s)$, the characteristic polynomial of cart 2 with cart 1 clamped at $x_1 = 0$. At such a root $s = z$, an exponential motion of cart 2 can balance an applied force on cart 1 through $U = -H(z)X_2$, while $X_1 = 0$. For $b = 0$, this is the familiar tuned-absorber cancellation at the real forcing frequency $\sqrt{(k_2 + k_3)/m_2}$; for $b > 0$, the zeros lie off the imaginary axis and do not give exact cancellation of a sustained sinusoidal input.

The numerator of X_2/U is the coupling force per unit displacement of cart 1, $bs + k_2$, which is only first order. For $b > 0$ and an exponential motion at $s = -k_2/b$, the coupling spring and damper forces cancel, so cart 2 can remain still.

This also explains the generic relative degrees: force acts directly on cart 1's acceleration, giving relative degree two, while transmission through the damper to cart 2 gives relative degree three for $b > 0$. If $b = 0$ and $k_2 > 0$, X_2/U has a constant numerator, no finite zeros, and relative degree four. Exceptional common numerator/denominator factors must be canceled before counting transfer-function zeros.

(c) Numerical modes

For the specified parameters,

$$A_1 = A_2 = s^2 + 0.5s + 15, \quad H = 0.5s + 5.$$

Using a difference of squares,

$$\Delta(s) = (s^2 + 10)(s^2 + s + 20) = s^4 + s^3 + 30s^2 + 10s + 200.$$

Comparison with $s^2 + 2\zeta\omega_n s + \omega_n^2$ gives:

Mode	Mode shape	Natural frequency ω_n	Damping ratio ζ	Poles
In phase	$(1, 1)^T$	$\sqrt{10} \approx 3.1623$ rad/s	0	$\pm i\sqrt{10}$
Out of phase	$(1, -1)^T$	$\sqrt{20} \approx 4.4721$ rad/s	$1/(2\sqrt{20}) \approx 0.1118$	$-1/2 \pm i\sqrt{79}/2$

The in-phase mode is undamped despite $b > 0$. Both damper ends move together, so it has no relative velocity and dissipates no energy. The unforced system is therefore not asymptotically stable.

(d) Nonzero initial displacements

For $x_i(0) = a_i$ and $\dot{x}_i(0) = 0$,

$$\mathcal{L}\{\ddot{x}_i\} = s^2 X_i - s a_i, \quad \mathcal{L}\{\dot{x}_1 - \dot{x}_2\} = s(X_1 - X_2) - (a_1 - a_2).$$

With $u = 0$, retaining these terms gives

$$\begin{bmatrix} A_1 & -H \\ -H & A_2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix},$$

where

$$R_1 = m_1 s a_1 + b(a_1 - a_2), \quad R_2 = m_2 s a_2 + b(a_2 - a_1).$$

The general answer is

$$X_1 = \frac{A_2 R_1 + H R_2}{\Delta}, \quad X_2 = \frac{H R_1 + A_1 R_2}{\Delta}.$$

For the parameters of (c), introduce

$$c = \frac{x_1 + x_2}{2}, \quad d = \frac{x_1 - x_2}{2}.$$

Adding and subtracting the unforced equations gives

$$\ddot{c} + 10c = 0, \quad \ddot{d} + \dot{d} + 20d = 0.$$

Therefore

$$C(s) = \frac{a_1 + a_2}{2} \frac{s}{s^2 + 10}, \quad D(s) = \frac{a_1 - a_2}{2} \frac{s + 1}{s^2 + s + 20},$$

and $X_1 = C + D$, $X_2 = C - D$. The three cases simplify to:

Case 1: $a_1 = a_2 = 1$ m.

$$X_1 = X_2 = \frac{s}{s^2 + 10}.$$

The entire out-of-phase factor cancels: this initial condition excites only the undamped in-phase mode.

Case 2: $a_1 = 1 \text{ m}$, $a_2 = -1 \text{ m}$.

$$X_1 = \frac{s+1}{s^2+s+20}, \quad X_2 = -\frac{s+1}{s^2+s+20}.$$

The in-phase factor cancels: this initial condition excites only the damped out-of-phase mode.

Case 3: $a_1 = 1 \text{ m}$, $a_2 = 0$.

$$\begin{aligned} X_1 &= \frac{1}{2} \frac{s}{s^2+10} + \frac{1}{2} \frac{s+1}{s^2+s+20} = \frac{s^3+s^2+15s+5}{(s^2+10)(s^2+s+20)}, \\ X_2 &= \frac{1}{2} \frac{s}{s^2+10} - \frac{1}{2} \frac{s+1}{s^2+s+20} = \frac{5s-5}{(s^2+10)(s^2+s+20)}. \end{aligned}$$

Both modes are excited. The cancellations in cases 1 and 2 are properties of the initial-condition response; they do not remove those modes from the underlying system.

(e) Closed-form responses and plots

Define

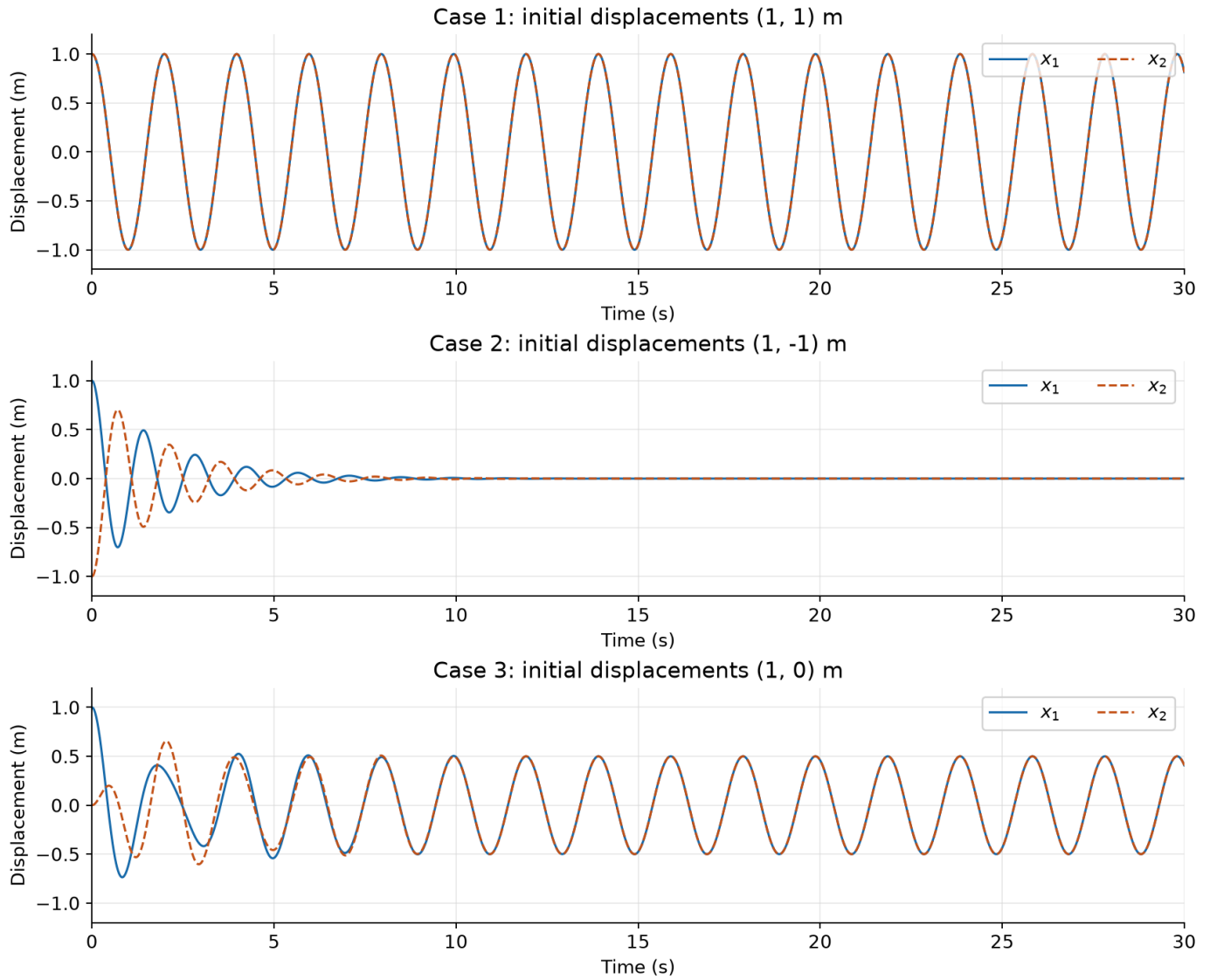
$$f(t) = \cos(\sqrt{10}t), \quad h(t) = e^{-t/2} \left[\cos\left(\frac{\sqrt{79}}{2}t\right) + \frac{1}{\sqrt{79}} \sin\left(\frac{\sqrt{79}}{2}t\right) \right].$$

These are the inverse transforms of $s/(s^2+10)$ and $(s+1)/(s^2+s+20)$, respectively. In particular,

$$\frac{s+1}{s^2+s+20} = \frac{s+1/2}{(s+1/2)^2+79/4} + \frac{1/2}{(s+1/2)^2+79/4}.$$

For $t \geq 0$, the displacements in meters are

Initial condition	$x_1(t)$	$x_2(t)$
Case 1: (1, 1) m	$f(t)$	$f(t)$
Case 2: (1, -1) m	$h(t)$	$-h(t)$
Case 3: (1, 0) m	$[f(t) + h(t)]/2$	$[f(t) - h(t)]/2$



In case 1, $x_1 = x_2$ and $\dot{x}_1 = \dot{x}_2$ for all time. The coupling spring is unextended, and the damper's force and dissipated power are both zero:

$$F_b = b(\dot{x}_2 - \dot{x}_1) = 0, \quad P_{\text{diss}} = b(\dot{x}_2 - \dot{x}_1)^2 = 0.$$

The outer springs exchange energy with the carts' kinetic energy indefinitely.

Superposition follows from

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

The case 3 trajectory is exactly half the case 1 trajectory plus half the case 2 trajectory, component by component. As $h(t)$ decays, both carts approach the same persistent oscillation $\frac{1}{2} \cos(\sqrt{10}t)$; they do not converge to rest.

(f) Changing damping and then breaking mass symmetry

Equal masses, with $b = 0, 2, 10 \text{ N}\cdot\text{s/m}$.

Keeping the other parameters at their values in (c), the modal equations become

$$\ddot{c} + 10c = 0, \quad \ddot{d} + 2b\dot{d} + 20d = 0,$$

where the numerical coefficients use SI units and $m_1 = m_2 = 1 \text{ kg}$. Thus

$$\Delta_b(s) = (s^2 + 10)(s^2 + 2bs + 20), \quad \zeta_d = \frac{b}{\sqrt{20}}.$$

For case 1, $x_1 = x_2 = f(t)$ for all three choices of b .

For case 3,

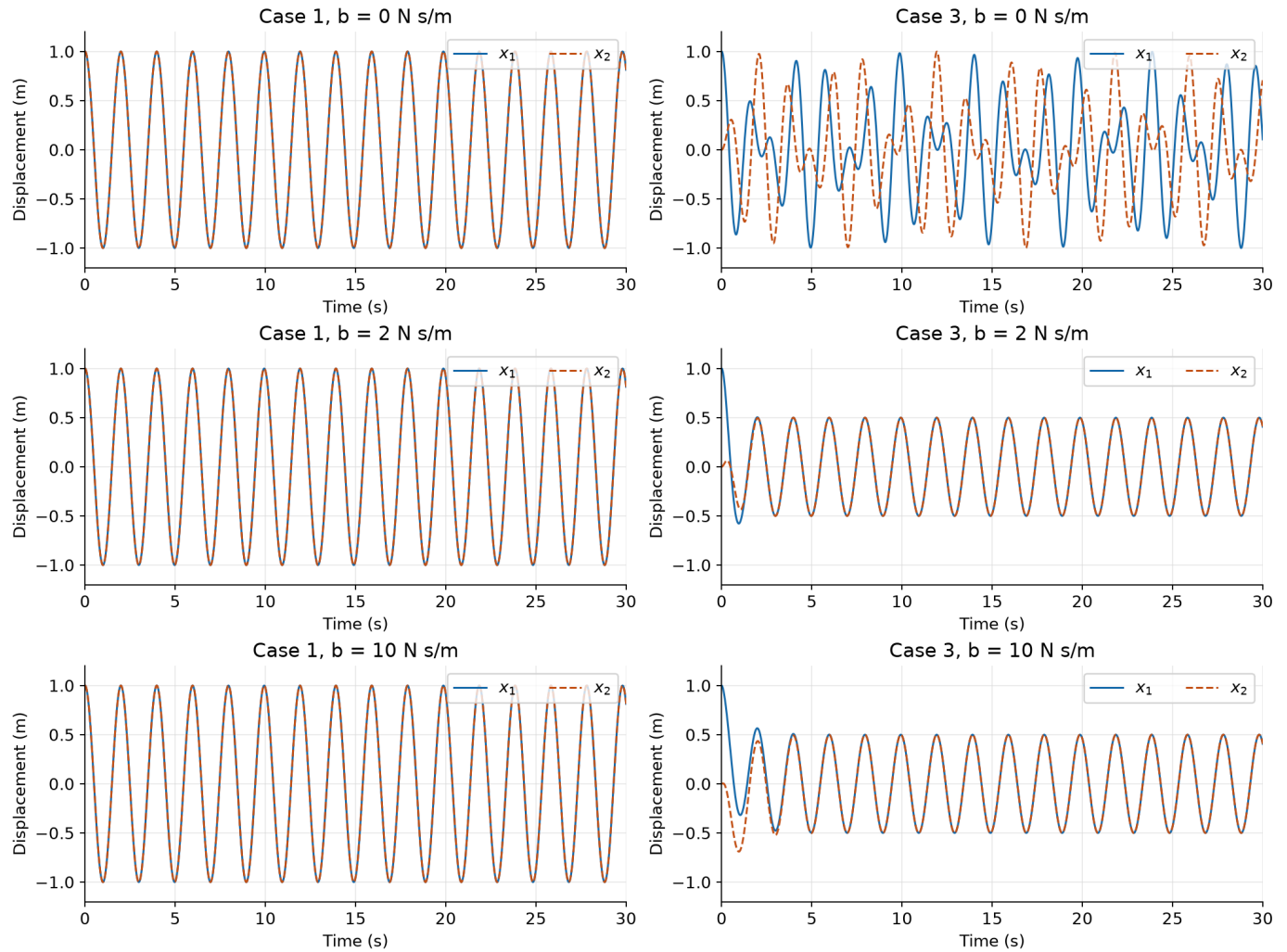
$$x_1 = \frac{f + h_b}{2}, \quad x_2 = \frac{f - h_b}{2},$$

where $h_b(0) = 1$, $\dot{h}_b(0) = 0$, and the exact functions are

$$\begin{aligned} h_0(t) &= \cos(\sqrt{20}t), \\ h_2(t) &= e^{-2t} \left[\cos(4t) + \frac{1}{2} \sin(4t) \right], \\ h_{10}(t) &= e^{-10t} \left[\cosh(4\sqrt{5}t) + \frac{\sqrt{5}}{2} \sinh(4\sqrt{5}t) \right]. \end{aligned}$$

For stable numerical evaluation of h_{10} , one may equivalently use

$$h_{10}(t) = \frac{2 + \sqrt{5}}{4} e^{(-10+4\sqrt{5})t} + \frac{2 - \sqrt{5}}{4} e^{(-10-4\sqrt{5})t}.$$



b (N·s/m)	Out-of-phase response	Effect on case 3
0	Undamped; poles $\pm i\sqrt{20}$	Both frequencies persist, producing two-frequency motion with a slow envelope and energy exchange between the carts.
2	Underdamped; $\zeta_d = 1/\sqrt{5} \approx 0.4472$, poles $-2 \pm 4i$	The relative oscillation decays faster than for $b = 0.5$; the common oscillation persists.
10	Overdamped; $\zeta_d = \sqrt{5}$, poles $-10 \pm 4\sqrt{5} \approx -1.0557, -18.9443$	The relative displacement returns without oscillation, but its slow tail decays more slowly than the $b = 2$ envelope. The common oscillation still persists.

Critical damping of the relative mode occurs at $b = \sqrt{20} \approx 4.4721$ N·s/m. Increasing b beyond this does not continually speed up the response.

Unequal masses: $m_2 = 3$ kg and $b = 0.5$ N·s/m.

Here

$$A_1 = s^2 + 0.5s + 15, \quad A_2 = 3s^2 + 0.5s + 15, \quad H = 0.5s + 5,$$

$$\Delta_3(s) = 3s^4 + 2s^3 + 60s^2 + 10s + 200.$$

The initial-condition transforms are

$$\begin{array}{ll} \text{Case 1:} & X_1 = \frac{3s^3 + 2s^2 + 30s}{\Delta_3(s)}, \quad X_2 = \frac{3s^3 + 2s^2 + 50s}{\Delta_3(s)}, \\ \text{Case 3:} & X_1 = \frac{3s^3 + 2s^2 + 15s + 5}{\Delta_3(s)}, \quad X_2 = \frac{5s - 5}{\Delta_3(s)}. \end{array}$$

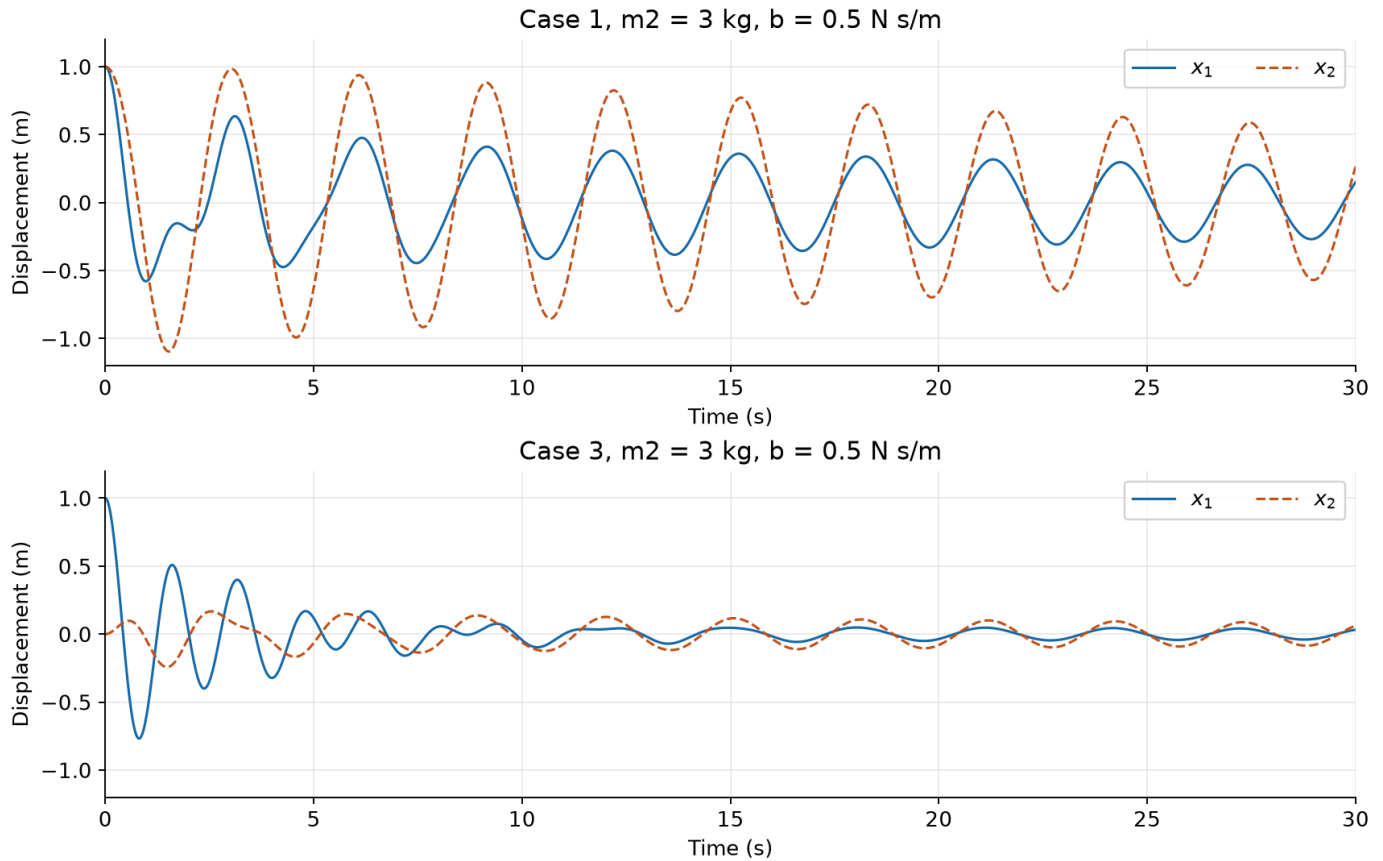
The poles, common to both cases, are approximately

$$s = -0.02213 \pm 2.05818i, \quad s = -0.31120 \pm 3.95462i.$$

An exact time-domain specification for each numerator $N_i(s)$ above is

$$x_i(t) = \sum_{\Delta_3(p)=0} \frac{N_i(p)}{\Delta_3'(p)} e^{pt},$$

with conjugate terms combined to give a real response. The plots evaluate the same responses numerically.



Both modes now decay in both cases. The slow pair has a decay time constant of about $1/0.02213 \approx 45.2$ s, so substantial motion can remain at $t = 30$ s. In case 1 the carts no longer move together; in case 3 there is no persistent common oscillation after the transients.

The reason is that equal displacement no longer produces equal acceleration. For case 1 at $t = 0$,

$$\ddot{x}_1(0) = -10 \text{ m/s}^2, \quad \ddot{x}_2(0) = -\frac{10}{3} \text{ m/s}^2.$$

Although the damper initially has zero force, relative velocity soon develops and it begins dissipating energy. More generally, a nonzero common motion $x_1 = x_2 = x$ would have to satisfy both $\ddot{x} + 10x = 0$ and $3\ddot{x} + 10x = 0$, which is impossible unless $x = 0$.

The mechanical energy satisfies

$$E = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 + \frac{1}{2}k_1x_1^2 + \frac{1}{2}k_2(x_1 - x_2)^2 + \frac{1}{2}k_3x_2^2,$$

$$\dot{E} = -b(\dot{x}_1 - \dot{x}_2)^2 \leq 0 \quad (u = 0).$$

For the unequal masses and fixed equal outer springs, no nonzero normal mode can avoid relative velocity. This removes the undamped mode. More generally, for $b > 0$, an undamped oscillatory mode must have $\dot{x}_1 = \dot{x}_2$, hence equal displacement amplitudes $x_1 = x_2 = x$. Substituting into the unforced equations gives

$$m_1\ddot{x} + k_1x = 0, \quad m_2\ddot{x} + k_3x = 0.$$

A nonzero common oscillation therefore exists precisely when $k_1/m_1 = k_3/m_2$; equality of masses and outer springs is one way to satisfy that condition. Linearity and superposition still hold after changing the mass, but the simple in-phase/out-of-phase modal separation does not.

(g) Unit-step steady state for $m_2 = 3$ kg

With zero initial conditions and a unit step force, $U(s) = 1/s$, so

$$X_1(s) = \frac{3s^2 + 0.5s + 15}{s\Delta_3(s)}, \quad X_2(s) = \frac{0.5s + 5}{s\Delta_3(s)}.$$

The poles of $sX_i(s)$ are strictly in the left half-plane, as verified in (f), so the final value theorem applies:

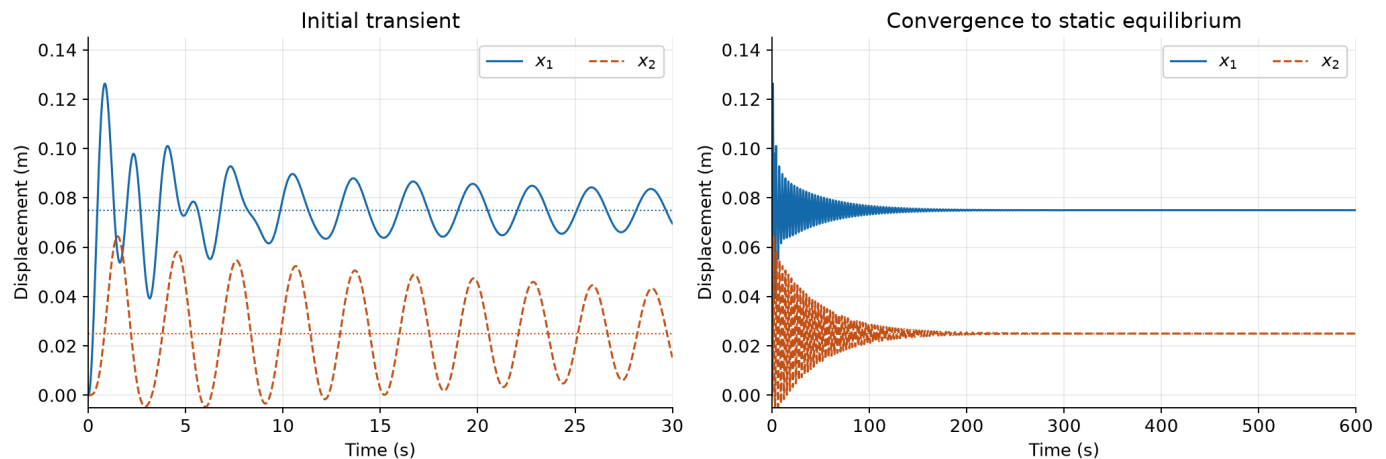
$$x_1(\infty) = \lim_{s \rightarrow 0} sX_1(s) = \frac{15}{200} = 0.075 \text{ m},$$

$$x_2(\infty) = \lim_{s \rightarrow 0} sX_2(s) = \frac{5}{200} = 0.025 \text{ m}.$$

At static equilibrium all velocities and accelerations vanish. The force balances reduce to

$$15x_1 - 5x_2 = 1, \quad -5x_1 + 15x_2 = 0.$$

The second equation gives $x_2 = x_1/3$; substitution into the first gives $x_1 = 3/40$ m and $x_2 = 1/40$ m, agreeing with the final value theorem. Mass and damping affect the transient but not this static balance.



Numerical integration through $t = 600$ s gives $x_1 \approx 0.075000$ m and $x_2 \approx 0.025000$ m, each within 10^{-6} m of its predicted limit. The longer time window is useful because of the slowly decaying pole pair.

For comparison, the final value theorem would not apply to the original equal-mass step responses: their undamped poles at $\pm i\sqrt{10}$ produce persistent oscillation, even though the same static force-balance solution exists.

Reproducing the figures and numerical checks

The accompanying [plotting script](#) generates every figure using NumPy, SciPy, and Matplotlib:

```
python ps1-answers-plots.py
```

It checks the closed-form responses in (e), every equal-mass damping variant in (f), the unequal-mass initial-condition transforms against numerical integration, and the step-response limits in (g).